

ulms-highgrade-pulm-mets-tp53-smarc b1-hla-a2-w9t4

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. Atorvastatin (mevalonate-pathway / protein geranylgeranylation blockade) repurposed for uterine leiomyosarcoma (repurposed drug)

Target: High-grade uterine leiomyosarcoma histology **Stage:** In vivo (animal) **Evidence strength:** Emerging

The only candidate whose causal chain was run in uterine LMS itself. Whether an antitumour dose is reachable in a person is untested.

First place goes to the only candidate here whose causal chain was run in her disease rather than borrowed from another sarcoma: uterine LMS cells and uterine LMS xenografts, with geranylgeranyl pyrophosphate add-back rescuing the phenotype and Rap1A/Rap1B unprenylation confirming the mechanism instead of assuming it. That is scarcer than it sounds. Only seven of the fourteen candidates ranked here were built in a smooth-muscle model at all, three of those carry lineage caveats, and the one published uterine LMS xenograft resource reports expression drift away from the parent tumour in three of ten models. Set against that, a cheap oral generic with an independent mechanism is worth a laboratory's time. What it is not is a therapy: the xenograft effect was roughly half of tumour growth rather than regression, it comes from one group and one paper, and the antitumour concentrations sit well above what lipid-lowering dosing reaches in plasma. The experiment that decides this is not another mouse study. It is whether that concentration is reachable in a person at a dose she could tolerate.

Mechanism. Atorvastatin blocks HMG-CoA reductase and drains the isoprenoid pool below mevalonate. Losing geranylgeranyl pyrophosphate leaves small GTPases such as Rap1A/Rap1B unprenylated, and the contractile smooth-muscle program collapses along with proliferation.

Translatability. High for the histology: uterine LMS lines and xenografts, with geranylgeranyl pyrophosphate rescue and Rap1A/Rap1B unprenylation confirming the mechanism rather than assuming it.

Developability. Closest to hand in this set. Atorvastatin is generic and oral, so the gap is dose: active concentrations sat far above lipid-lowering exposure.

Counter-productive MoA. Low (Mevalonate blockade slows proliferation, which could in principle blunt cycle-dependent cytotoxics given at the same time)

Key risks.

- The exposure gap is the whole question. Concentrations that killed cells in culture are far above what lipid-lowering dosing achieves in plasma
- One laboratory, one publication, no independent replication yet
- Statin myopathy and rhabdomyolysis risk would rise alongside the cytotoxic regimen she is scheduled to start
- The xenograft result was partial growth inhibition, not regression
- The cell lines are the same long-passaged pair that most uterine LMS work leans on, with known drift from primary tumour biology

Open questions.

- What plasma and intratumoral concentration does a tolerated oncology dose actually produce in a human?
- Does a second laboratory reproduce the xenograft effect, and does it hold in a model that has not drifted?
- How would a statin interact with doxorubicin and trabectedin, given overlapping muscle and liver toxicity?

Key references. pmid:42252304

2. SUV39H2 inhibitor OTS186935 plus a PARP inhibitor (combination)

Target: High-grade uterine leiomyosarcoma histology; DNA-repair capacity still unmeasured **Stage:** In vivo (animal) **Evidence strength:** Emerging

An HRD-independent route onto the repair axis, built in uterine LMS models but resting on a chemical probe nobody has taken toward a patient.

The interesting property here is that the combination manufactures a repair defect rather than requiring one. Her homologous-recombination status has never been tested on tissue, and the plasma panel does not settle it, so every strategy gated on an HRD result is gated on a test she has not had. That matters because this dossier's strongest lineage-matched signal sits on the DNA-repair axis, and it came from PARP1-selective AZD5305 in biallelic BRCA2-deleted uterine LMS models including a paired post-resistance one: a genotype nobody has looked for in her. An HRD-independent way onto the same axis is therefore worth something. The target is anchored in the right tissue, with SUV39H2 protein higher in uterine LMS than in matched myometrium or leiomyoma, and the synthetic lethality was shown in uterine LMS lines and then in xenografts. It sits second rather than first because OTS186935 is an academic probe with no clinical-grade successor, the combination's haematologic toxicity is entirely uncharacterised, and the experiments never formally separated whether the benefit still needs an underlying repair defect.

Mechanism. SUV39H2 methylates H3K9 and marks H2AX at K134, which efficient gamma-H2AX deposition and double-strand-break repair depend on. Inhibiting it leaves breaks poorly signalled; adding PARP inhibition removes the single-strand route that would otherwise make those lesions survivable.

Translatability. Moderate: uterine LMS lines plus xenograft, with SUV39H2 protein higher in uterine LMS than in matched myometrium, though both lines are long-passaged.

Developability. Far off. OTS186935 is an academic probe with no clinical-grade successor, so a SUV39H2 chemotype would need carrying through medicinal chemistry first.

Counter-productive MoA. Moderate (Layering PARP inhibition onto broad methyltransferase blockade could stack myelosuppression and force both doses down)

Key risks.

- H3K9 methyltransferase inhibition has broad chromatin consequences and no clinical safety precedent in any indication
- Stacking a PARP inhibitor onto it could compound myelosuppression in ways nobody has measured
- Whether the benefit still requires an underlying repair defect was never separated experimentally
- Both uterine LMS lines used are long-passaged and drift from primary tumour biology

Open questions.

- Does the combination still work in an HR-proficient background, or is it quietly selecting for the defect she has not been tested for?
- Is there a second SUV39H2 chemotype anywhere that could carry this past probe stage?
- The screener flagged a clinical-track item on this same axis while this scan was maturing: NCT05711615, DNA-PK inhibition with low-dose doxorubicin, second line or later and capped at 360 mg/m² of prior anthracycline. That belongs to the trial screen, not to this page.

Key references. pmid:41402789

3. Aurora A kinase inhibition selected by the uLMS centrosome/spindle transcriptional program (MK-5108) (small molecule)

Target: High-grade uterine leiomyosarcoma histology; TP53-deficient mitotic background **Stage:** In vivo (animal)

Evidence strength: Moderate

Right model, old data. The uterine LMS selection rationale is the novel part; the target class itself has already disappointed in patients elsewhere.

Expression profiling across myometrium, leiomyoma and leiomyosarcoma pointed at the mitotic apparatus as the most consistently deranged program in this histology, and the orthotopic experiment is what earns the rank: oral dosing shrank implants in a uterine model rather than a flank tumour. A TP53-null background removes the checkpoint that would let a cell survive a botched mitosis, though note that her TP53 call is plasma-only and carries an explicit clonal-haematopoiesis caveat, so the premise is a population expectation for uterine LMS rather than a measured feature of her tumour. Three things hold it here. The supporting work is from 2012 and nobody has repeated it in a modern uterine LMS model. MK-5108 itself was abandoned after one early-phase study. And it did not reliably synergise with gemcitabine or docetaxel, which makes it a single-agent hypothesis rather than a way to make chemotherapy work harder. Substituting a current Aurora-A-selective molecule would move the whole question onto the clinical track.

Mechanism. MK-5108 is a highly Aurora-A-selective ATP-competitive inhibitor. Over half the most overexpressed transcripts in uterine LMS govern centrosome and spindle function, so blocking Aurora A pushes cells with an already fragile mitotic apparatus into failed division.

Translatability. Moderate: an orthotopic uterine model rather than a flank tumour, but the work is from 2012 and no modern uterine LMS model has repeated it.

Developability. Awkward. MK-5108 is effectively a tool compound now, and swapping in a current Aurora-A-selective molecule moves the idea onto the clinical track.

Counter-productive MoA. Moderate (Arresting cells before mitosis could shield them from mitotic-phase cytotoxics such as docetaxel)

Key risks.

- Aurora A inhibitors carry class neutropenia and mucositis, and the broader clinical development of this target class largely disappointed

- No synergy with gemcitabine or docetaxel was shown, which closes off the obvious combination strategies
- The dependency was inferred from expression, not from a functional screen in uterine LMS
- Her TP53 status rests on a plasma call with a clonal-haematopoiesis caveat, so the mitotic-checkpoint premise is unconfirmed

Open questions.

- Does the orthotopic result reproduce with a current Aurora-A-selective molecule in a modern uterine LMS model?
- Is the dependency real or an artefact of reading it off expression rather than a functional screen?

Key references. pmid:22535157

4. Glyoxalase-1 inhibition (BBGC) to potentiate trabectedin in soft-tissue sarcoma (combination)

Target: High-grade uterine leiomyosarcoma histology; planned trabectedin-containing first line **Stage:** In vivo (animal) **Evidence strength:** Emerging

Aimed squarely at the regimen she is scheduled to receive, with in-vivo evidence that sits in the wrong sarcoma.

This is the candidate closest to the treatment she is actually scheduled to receive. Doxorubicin plus trabectedin is planned but not started, and trabectedin's single-agent activity in leiomyosarcoma is modest enough that a sensitiser is a reasonable thing to want. The screen that nominated glyoxalase-1 used sarcoma expression data and a panel of lines that included leiomyosarcoma, and the potentiation held in vivo. The catch is which sarcoma: the in-vivo arm ran in a fibrosarcoma orthotopic model, not a leiomyosarcoma one, so the lineage match stops at the cell-line panel. Trabectedin's hepatotoxicity and myelosuppression also leave little headroom for anything that raises its effective dose, and the source paper carries industry co-authorship from the trabectedin manufacturer, which is worth the usual caution when reading a combination framing.

Mechanism. BBGC is a cell-permeable diester prodrug that inhibits glyoxalase-1, the enzyme clearing methylglyoxal. Letting methylglyoxal accumulate raises dicarbonyl stress and adduct burden, which appears to lower the threshold at which trabectedin's minor-groove adducts turn lethal.

Translatability. Moderate: leiomyosarcoma lines were in the screening panel, but the in-vivo confirmation ran in a fibrosarcoma orthotopic model.

Developability. Far off. BBGC has no clinical-grade successor and no published pharmacokinetics, and a trabectedin pairing needs its own toxicology package.

Counter-productive MoA. Moderate (Added dicarbonyl stress compounds trabectedin hepatotoxicity and myelosuppression, which could force the trabectedin dose down)

Key risks.

- In-vivo confirmation is in fibrosarcoma, and cell-line potentiation is a weaker claim than an in-vivo one
- Glyoxalase-1 is broadly expressed, and what systemic methylglyoxal accumulation does to normal organs is unclear

- Trabectedin hepatotoxicity and myelosuppression cap how much sensitisation is usable
- Industry co-authorship from the trabectedin manufacturer warrants caution about how the combination is framed

Open questions.

- Does the potentiation hold in a leiomyosarcoma model in vivo, or only in the cell-line panel?
- Is there any dose of a glyoxalase-1 inhibitor that adds effect without eating trabectedin's narrow liver and marrow margin?

Key references. pmid:39198616

5. MIF-CD74/CD44 axis blockade (4-IPP tool compound; ibudilast as a repurposing handle) (repurposed drug)

Target: Immune-suppressed, T-cell-excluded metastatic microenvironment **Stage:** In vivo (animal) **Evidence strength:** Emerging

Names why checkpoint blockade fails in this disease, then tests the fix in a different sarcoma. The uterine LMS evidence is descriptive only.

Checkpoint blockade has failed repeatedly in this disease, so a mechanism aimed at why it fails is more interesting than another checkpoint agent. The single-cell map of metastatic uterine LMS names this circuit alongside exhausted CD8 T cells, M2 macrophages and immature neutrophils, which means the target was read off the right disease and the right compartment. Her lesions were described radiologically as infiltrative and possibly inflammatory, a radiology impression rather than a pathological finding, but enough to make the myeloid compartment worth interrogating. What holds it at fifth is the shape of the evidence: the uterine LMS work identifies the circuit and never perturbs it, while every therapeutic experiment behind the idea was done in osteosarcoma. The two available molecules are also awkward, a covalent tool compound with poor drug-like properties and an old phosphodiesterase inhibitor whose MIF affinity is modest.

Mechanism. Tumour-derived MIF signals through CD74 with CD44 as co-receptor, holding macrophages in an M2-like state and blunting CD8 T-cell function. 4-IPP covalently modifies the tautomerase site and destabilises the protein; ibudilast binds the same pocket allosterically.

Translatability. Moderate: the circuit was mapped in metastatic uterine LMS, but every perturbation experiment behind it was run in osteosarcoma.

Developability. Mixed. 4-IPP is a covalent tool compound; ibudilast is approved elsewhere and reaches patients faster but binds MIF weakly and is untested in sarcoma.

Counter-productive MoA. Moderate (Trabectedin already depletes tumour macrophages, so a repolarising agent could find the compartment gone)

Key risks.

- The uterine LMS evidence is descriptive, from one treatment-naive patient's metastases, with no perturbation experiment in this histology
- MIF has real roles in innate immunity, and sustained blockade carries an unquantified infection risk
- 4-IPP's covalent chemistry may hit other targets at the concentrations used

- Ibudilast at approved dosing may never reach MIF-inhibitory exposure

Open questions.

- Does blocking the circuit do anything in a uterine LMS model, which nobody has tried?
- Can ibudilast reach MIF-inhibitory concentrations at any tolerated dose?
- How would this interact with trabectedin, which acts on the same myeloid compartment?

Key references. pmid:40948800; pmid:40846326; pmid:35060345

6. Hypomethylating plus HDAC-inhibitor priming to induce MAGE-A4 / NY-ESO-1 / PRAME before HLA-A*02:01-restricted T-cell redirection (combination)

Target: HLA-A*02:01 confirmed; cancer/testis antigen expression untested **Stage:** In vivo (animal) **Evidence strength:** Emerging

The only published way to reopen the A*02:01 axis if antigen staining comes back negative. Induced antigen is not what any programme screens on.

Her HLA-A02:01 typing is confirmed, which settles the harder half of eligibility for A02:01-restricted T-cell redirection. The missing half is antigen expression, and MAGE-A4 / NY-ESO-1 / PRAME staining has never been run on her tissue. The right order is unchanged: stain first, on the biopsy that already exists. This candidate matters only for the branch where that staining comes back negative, because it is the only published route that reopens the axis rather than closing it. Decitabine raises cancer/testis antigens and MHC class I in sarcoma lines, a demethylating pretreatment produced antigen-specific CD8 responses in osteosarcoma, and a 2026 preprint extends sequential decitabine then entinostat to poorly immunogenic murine sarcoma. Both priming drugs are old and available, so the obstacle is the schedule rather than the molecules. Two honest limits: no leiomyosarcoma model has been used for any of this, and induced antigen is not what any T-cell-receptor programme currently accepts for eligibility.

Mechanism. Cancer/testis antigen promoters are silenced by CpG methylation. Decitabine demethylation restores transcription while HDAC inhibition opens the matching chromatin, and the same treatment raises MHC class I and ICAM-1, so an invisible tumour becomes both antigen-positive and better able to present.

Translatability. Moderate: decitabine raised antigen and MHC class I in sarcoma lines and a murine sarcoma, but no leiomyosarcoma model has been tested.

Developability. Both priming drugs are old and available. What is missing is a schedule: how much induction, how long it holds, alongside which product.

Counter-productive MoA. Moderate (Priming raises antigen on normal tissue as well, widening the target an affinity-enhanced receptor would attack)

Key risks.

- Antigen induction is patchy and may never reach the threshold a screening assay demands
- Hypomethylating agents raise antigen on normal tissue too, which could widen on-target off-tumour toxicity from

an affinity-enhanced receptor

- Myelosuppression from priming compounds the lymphodepletion that cell therapy already requires
- No leiomyosarcoma model sits anywhere behind this
- Induced expression is not what any current programme screens on, so eligibility would not follow even if it worked

Open questions.

- Does priming induce antigen in leiomyosarcoma specifically, which nobody has tested?
- How long does induced expression persist, and does it overlap the window a cell-therapy infusion needs?
- Would any sponsor accept induced rather than constitutive expression for screening?

Key references. pmid:24584817; pmid:25301731; biorxiv:10.64898/2026.06.18.733244

7. Copper loading to block ATP7B-mediated platinum efflux in uterine leiomyosarcoma (repurposed drug)

Target: High-grade uterine leiomyosarcoma histology; platinum resistance **Stage:** In vivo (animal) **Evidence strength:** Emerging

Reopening platinum in this disease is a large claim resting on one cell line. The ATP7B expression finding is the part likely to survive.

Platinum sits outside standard uterine LMS therapy because the disease does not respond to it, and this work offers a mechanistic reason rather than a shrug. Suppressing ATP7B cut the cisplatin IC50 several-fold and turned a negligible xenograft response into a substantial one, with copper pretreatment reproducing the effect pharmacologically. The lineage is right, which is why it sits ahead of the mechanistically prettier telomere candidates below. The rank stops here because the evidence and the intervention pull apart. Everything rests on one cell line and its isogenic derivative from a single group, with no confirmation in a patient-derived model. ATP7B expression in her tumour has never been measured. And copper loading is not a therapeutic strategy anyone has developed: the clinical precedent runs the other way, toward chelation, and copper toxicity has a narrow window. The durable part of this is the ATP7B expression finding; the copper intervention is the weak half.

Mechanism. ATP7B is a copper-transporting ATPase that also exports platinum drugs, and uterine LMS cells express it strongly. Preloading with copper occupies the transporter, platinum accumulates instead of being pumped out, and cisplatin regains cytotoxicity.

Translatability. Moderate for the target, low for the intervention: right histology and a large xenograft effect, but one cell line from one group.

Developability. Poor. Systemic copper loading is nobody's development plan, no selective ATP7B inhibitor exists, and clinical precedent runs the opposite way, toward chelation.

Counter-productive MoA. Moderate (Copper is pro-angiogenic, so loading could feed the biology that chelation trials set out to starve)

Key risks.

- Copper toxicity, hepatic and neurologic, is well documented and the therapeutic window is narrow

- One cell line and its isogenic derivative from a single group, with no patient-derived confirmation
- It presupposes a platinum backbone that is not standard in this disease
- ATP7B expression in her tumour is unmeasured and would have to be shown before any of this applied to her

Open questions.

- Does the effect reproduce in a patient-derived uterine LMS model, or is it a property of one line?
- Could a selective ATP7B inhibitor do this without systemic copper loading?

Key references. pmid:31351063

8. FANCM ablation in ALT-positive tumours (FANCM-RMI interaction inhibitors; FANCM antisense oligonucleotides) (oligonucleotide)

Target: Unmeasured molecular landscape (ALT / ATRX / HRD status all untested) **Stage:** In vivo (animal)

Evidence strength: Moderate

The cleanest genetics on the telomere axis, in the wrong sarcoma. A related node already failed in leiomyosarcoma despite confirmed target engagement.

This has the cleanest genetics on the telomere axis: CRISPR validation of the dependency, a screened antisense gapmer with in-vivo activity, and a published assay for a second modality against the FANCM-RMI interface. It still ranks below every lineage-matched candidate above, for a reason this case has already learned the hard way. The telomere thesis has been tested in her histology and failed. An ATRX-mutant leiomyosarcoma cohort progressed within four months on an ATR inhibitor despite biopsy-proven target engagement, and on-treatment SLFN11 rather than ATRX status tracked with whatever benefit there was. Different node, same axis, and that result should temper how a reader hears any claim of telomere-selective killing. Two further gaps. The in-vivo proof of concept is an ALT-positive liposarcoma, not a leiomyosarcoma. And her own ALT and ATRX status has never been measured, so this is a bet on a population prevalence of roughly 59% in leiomyosarcoma, not a patient-level rationale.

Mechanism. Cells using alternative lengthening of telomeres run continuous replication stress there and depend on FANCM's branchpoint translocase activity to survive it. Removing FANCM, by disrupting the FANCM-RMI interaction or knocking the transcript down, lets that stress run away.

Translatability. Low, cross-tumour only: an antisense gapmer in ALT-positive liposarcoma. No leiomyosarcoma model has been tested and her ALT status is unmeasured.

Developability. Far off. An academic gapmer with in-vivo activity and a published screening assay, but no IND-enabling work, and lung delivery adds its own problem.

Counter-productive MoA. Moderate (Driving telomeric replication stress in normal cells risks the instability that Fanconi-pathway loss produces)

Key risks.

- An ATR inhibitor already failed in ATRX-mutant leiomyosarcoma despite proven target engagement, which is a cautionary precedent for telomere-selectivity claims generally
- In-vivo proof of concept is in liposarcoma, not leiomyosarcoma

- Her ALT status is unmeasured, so a telomere assay on tissue would have to come first
- FANCM loss drives chromosome instability in normal cells and a Fanconi-like phenotype

Open questions.

- Does a leiomyosarcoma model show the dependency at all?
- Given the ATR inhibitor result, is telomere-maintenance status still worth assaying on her tissue?
- Does SLFN11 rather than ALT mark who benefits on this axis?

Key references. pmid:28673972; pmid:38886039; pmid:40125273

9. MYOCD / MRTF-SRF smooth-muscle lineage transcription-factor dependency (conceptual strategy)

Target: High-grade uterine leiomyosarcoma histology (smooth-muscle lineage) **Stage:** Conceptual **Evidence strength:** Speculative

The most under-explored idea on this list rather than the best supported. The first question is whether uterine LMS depends on MYOCD at all.

Lineage-survival transcription factors have been a productive target class in other histologies, and this disease has an obvious candidate nobody has drugged. MYOCD is amplified and overexpressed in most retroperitoneal leiomyosarcomas, knockdown in an amplified line stripped smooth-muscle differentiation and slowed growth, and later genomic subtyping showed leiomyosarcoma splitting along lineage lines that track survival. That is enough to make differentiation state look like it is doing real work in this disease. It is not enough to make this a drug proposal. The functional work is retroperitoneal rather than uterine, and MYOCD amplification is less frequent at her site. No MYOCD-directed molecule exists, and the MRTF/SRF probes were built for fibrosis with cancer pharmacology never worked out. The first experiment is not a therapeutic one: it is whether uterine LMS models depend on MYOCD at all.

Mechanism. Myocardin is the smooth-muscle-specific coactivator of serum response factor and sets the contractile differentiation program defining this lineage. Removing it, genetically or in principle by degradation or blockade of the MRTF/SRF interface, collapses that program.

Translatability. Low: the functional work is retroperitoneal leiomyosarcoma and one amplified line. Whether uterine LMS carries the same dependency is untested.

Developability. Nothing exists. No MYOCD-directed molecule, and the MRTF/SRF probes were built for fibrosis with their cancer pharmacology never worked out.

Counter-productive MoA. Moderate (Stripping the differentiation program could select for a dedifferentiated state that behaves worse)

Key risks.

- The functional work is in retroperitoneal leiomyosarcoma, and MYOCD amplification is less frequent at the uterine site
- Smooth muscle is essential tissue in the vasculature, gut and airway, so on-target toxicity could be severe
- MRTF/SRF probes have modest potency and poorly characterised selectivity

- This is a target-validation proposal, not a drug proposal

Open questions.

- Do uterine LMS models show the dependency that retroperitoneal ones did?
- Is the MRTF/SRF interface tractable enough to build a cancer-grade molecule against?

Key references. pmid:19276386; pmid:34301934; pmid:24706986

10. SMARCAL1 annealing-helicase inhibition in ATRX-deficient, ALT-positive tumours (conceptual strategy)

Target: ALT with ATRX loss (both unmeasured in this patient) **Stage:** In vivo (animal) **Evidence strength:** Emerging

A different logic on the telomere axis: kill by overdriving it. No inhibitor exists and both selection markers are unmeasured here.

The logic is the interesting part: kill by pushing telomere recombination past what the cell can absorb, rather than by starving it. That is a different failure mode from the one an ATR inhibitor tried and, in principle, might not inherit the selectivity problem that sank it in an ATRX-mutant leiomyosarcoma cohort. It is also the newest and most specific dependency on this axis, drawn from dependency-map data and then confirmed in patient-derived models and intracranial xenografts. Everything else argues for holding it here. Every experiment was done in glioma, with no sarcoma model touched. Inducible knockdown is a gentler perturbation than a drug and tends to flatter the therapeutic window. Germline SMARCAL1 loss causes a multisystem developmental syndrome, which hints at what systemic inhibition might cost. And there is no inhibitor of any kind, which is the whole obstacle. Both selection markers, ALT and ATRX, are also unmeasured in her.

Mechanism. SMARCAL1 is an annealing helicase that remodels stalled replication forks and localises to ALT-associated PML bodies. Depleting it hyperactivates the ALT phenotype rather than shutting it down, producing G2 breaks and death by mitotic catastrophe.

Translatability. Low, cross-tumour only: every experiment is in glioma, and inducible knockdown is a gentler perturbation than any drug would be.

Developability. None yet. Genetic depletion only, no chemical matter, and a sarcoma-relevant version needs the dependency screen re-run before discovery could start.

Counter-productive MoA. Moderate (Hyperactivating telomere recombination could raise genomic instability in surviving cells rather than killing them)

Key risks.

- Every experiment is in glioma; no sarcoma model has been tested
- Inducible knockdown often overstates a therapeutic window that a small molecule would not reproduce
- Germline SMARCAL1 loss causes a multisystem developmental syndrome, hinting at real on-target toxicity
- Requires both ALT and ATRX status, neither of which has been measured here

Open questions.

- Does the dependency-map analysis hold when re-run on sarcoma lines?
- Would a small molecule reproduce what inducible knockdown showed, or is the window an artefact of the perturbation?

Key references. pmid:41520142

11. CIP2A-TOPBP1 interface disruption in HR-deficient leiomyosarcoma (conceptual strategy)

Target: Homologous-recombination deficiency (unmeasured; roughly 35% of leiomyosarcomas carry HRD signatures) **Stage:** In vitro **Evidence strength:** Emerging

An undrugged node on the axis where this histology's strongest signal sits, gated on a repair-deficiency result she has never had.

Around a third of leiomyosarcomas carry repair-deficiency signatures, and the repair axis is where this dossier's strongest lineage-matched signal sits. What distinguishes this node from PARP, POLQ and USP1 is that it remains genuinely undrugged, which is why it belongs on a horizon scan rather than a trial screen. The mechanism is mitotic rather than replicative, so it should not share cross-resistance with PARP inhibition, which would matter if she ever receives one. Against that: every experiment is in BRCA-mutant breast and ovarian contexts with nothing in sarcoma, and it is a protein-protein interface with no reported chemistry, so even an early campaign is years from a testable molecule. The selection marker is also hypothetical here. Her homologous-recombination status has never been assessed on tissue, and the lineage-matched PARP result this axis leans on came from biallelic BRCA2-deleted models, a state nobody has looked for in her.

Mechanism. CIP2A and TOPBP1 tether shattered chromosome fragments through mitosis and support mitotic DNA synthesis and microhomology-mediated end joining. HR-deficient cells lean on that tethering to survive their own break burden.

Translatability. Low, cross-tumour only: BRCA-mutant breast and ovarian models with no sarcoma work, and her repair status has never been tested on tissue.

Developability. Far off. A protein-protein interface with no reported small molecule or degrader, and interface inhibitors of this class rarely convert.

Counter-productive MoA. Low (CIP2A also restrains PP2A, so blocking it may carry pleiotropic effects beyond the intended repair defect)

Key risks.

- All evidence comes from BRCA-mutant breast and ovarian contexts, none from sarcoma
- Requires a repair defect that is unmeasured here and would need tissue testing first
- Interface inhibitors of this class have a poor historical hit rate
- CIP2A also restrains PP2A, so on-target effects are likely to be pleiotropic

Open questions.

- Is the dependency present in HR-deficient leiomyosarcoma models, which do not currently exist in usable form?
- Is the interface druggable at all, by small molecule or degrader?

Key references. pmid:35121901; pmid:41309571

12. ADAR1 inhibition in BRCA1/2-mutant or HR-deficient tumours (conceptual strategy)

Target: Homologous-recombination deficiency (unmeasured); immune-cold microenvironment **Stage:** In vitro

Evidence strength: Emerging

Two of her open questions point the same way, but the dependency was defined in BRCA-mutant breast and ovarian cells, nowhere near sarcoma.

Two of this case's open questions point the same way, which is what puts the idea on the list: an unmeasured repair-deficiency fraction that is substantial in leiomyosarcoma, and a microenvironment that single-cell work describes as exhausted and myeloid-dominated. A dependency here would hit the tumour cell directly and, as a side effect, raise the interferon signal a T-cell-excluded tumour lacks. Whether that second effect helps or simply drives adaptive resistance markers up is unresolved and would be the first thing worth testing. It ranks low because nothing about this has been examined in sarcoma, the dependency was defined in BRCA-mutant breast and ovarian cells, the selection biomarker is unmeasured in her, and the inhibitors are early discovery-stage with nothing publicly in IND-enabling work.

Mechanism. ADAR1 edits double-stranded RNA so cytosolic sensors do not read it as viral. In BRCA1/2-mutant cells the baseline interferon tone is already raised, and removing ADAR1 tips it into a lethal cell-intrinsic interferon response.

Translatability. Low, cross-tumour only: BRCA1/2-mutant breast and ovarian lines, nothing in sarcoma, and the repair defect it selects on is unmeasured here.

Developability. Early discovery across several groups. No ADAR1 inhibitor or degrader is publicly in IND-enabling work, and no matched sarcoma model exists to test one.

Counter-productive MoA. Moderate (Interferon induction can raise PD-L1 and other adaptive resistance signals instead of warming the tumour)

Key risks.

- ADAR1 loss in normal tissue drives interferonopathy, so systemic inhibition may be poorly tolerated
- The dependency was defined in BRCA-mutant breast and ovarian contexts, not sarcoma
- Her repair status is unmeasured, so even the selection biomarker is hypothetical
- Interferon induction may raise adaptive resistance signals rather than warm the tumour

Open questions.

- Does interferon induction help here, or does it mostly raise adaptive resistance markers?
- Is the dependency present in HR-deficient sarcoma, which nobody has tested?

Key references. pmid:40730818

13. Deregulating BLM at ALT telomeres via the SLX4IP/FANCM brake

(conceptual strategy)

Target: Alternative lengthening of telomeres (unmeasured) **Stage:** In vitro **Evidence strength:** Speculative

Best run as a second arm of the FANCM experiment rather than its own programme. One paper and an unrefereed preprint stand behind it.

Both directions on this axis are lethal to ALT cells in principle, which is unusual enough to be worth testing empirically rather than arguing about. Practically it is a second node on the same pathway as the FANCM candidate, so a laboratory pursuing that one should fold this into the same experiment rather than run it as a separate programme. The support is thin by design of the evidence, not by choice: one peer-reviewed paper and an unrefereed preprint, no in-vivo work in any tumour type, no leiomyosarcoma model, and BLM chemical probes too weak and promiscuous to build on. It also inherits the caution the whole telomere branch carries in this case, where a related node already failed in leiomyosarcoma despite confirmed target engagement, and her telomere-maintenance status has never been measured.

Mechanism. BLM helicase drives telomere recombination in ALT but has to be restrained, with SLX4IP acting in parallel to FANCM as that brake. Releasing the brake, or blocking BLM outright with a probe, pushes telomeres past the recombination load they tolerate.

Translatability. Low, cross-tumour only: osteosarcoma and fibroblast lines, no in-vivo work anywhere, and BLM probes that are weak and promiscuous.

Developability. Research reagents only. SLX4IP is a scaffold protein without an obvious pocket, and the available BLM probes are too unselective to build on.

Counter-productive MoA. Moderate (Systemic BLM inhibition mimics Bloom syndrome biology, including its cancer predisposition)

Key risks.

- BLM loss causes a cancer-predisposition syndrome, so systemic inhibition is a hard proposition
- One peer-reviewed paper plus an unrefereed preprint
- No in-vivo data at all, in any tumour type
- Depends on a telomere-maintenance status that has not been measured here

Open questions.

- Does either direction on this axis do anything in vivo?
- Is there a BLM chemotype selective enough to test the hypothesis honestly?

Key references. pmid:31447390; biorxiv:10.1101/2025.05.28.656696

14. SMARCB1 G344Afs*13 neo-open-reading-frame peptide as a personalised HLA-A*02:01 immunogen (conceptual strategy)

Target: HLA-A*02:01 confirmed; SMARCB1 frameshift (plasma-only, unconfirmed) **Stage:** Conceptual **Evidence strength:** Speculative

Listed because her two confirmed facts intersect here, not because evidence exists. A plasma-only heterozygous frameshift is a weak place to start.

This is listed because the two confirmed facts in her profile intersect here, not because evidence stands behind it. It ranks last, and the biomarker is why. The frameshift was called on plasma alone at an unstated variant fraction in a report whose other variants sat at or near the assay's limit of detection, it has never been confirmed on tissue, and it is heterozygous, so wild-type protein remains. A single plasma-only heterozygous call is a much weaker predictor than a confirmed biallelic one, and the prior in this histology is low: a series of 170 consecutive uterine smooth-muscle tumours retained SMARCB1 throughout. To be explicit about scope, this is not the SMARCB1-deficiency branch that leads to EZH2 inhibition. That branch is closed for this case on the same evidence. What is proposed here is narrower and purely immunological, and it is answerable cheaply. The frameshift yields only about thirteen novel residues before the stop, so the peptide space is small and may contain no A*02:01 binder at all, a question a binding predictor and a stabilisation assay could settle in an afternoon. Everything downstream still depends on the variant being tumour-derived, expressed, and surviving nonsense-mediated decay.

Mechanism. A frameshift produces a short stretch of amino acids found nowhere in the normal proteome before the premature stop. Those residues carry no thymic tolerance, so any that bind HLA-A*02:01 could in principle be targeted by a vaccine or a receptor.

Translatability. None yet: no model, no peptide, no binding data, and everything rests on a plasma-only variant that tissue testing has not confirmed.

Developability. Nothing exists. Confirm on tissue, show the transcript survives decay, then measure A*02:01 binding. Any one of those steps can end it.

Counter-productive MoA. Low (Immune pressure on a possibly subclonal variant could select antigen-negative outgrowth)

Key risks.

- The variant is plasma-only, heterozygous and unconfirmed on tissue, and SMARCB1 alteration is rare in uterine smooth-muscle tumours
- Nonsense-mediated decay may destroy the transcript before any peptide is made
- Thirteen novel residues is a small peptide space and may contain no A*02:01 binder
- A retained wild-type allele means this is an immunological target, not a dependency
- No manuscript, model or product addresses this idea; it is case-specific by construction

Open questions.

- Does the variant appear on tissue at all, and is the transcript expressed?
- Is there any A*02:01 binder in the thirteen novel residues? An afternoon of prediction and a stabilisation assay would answer it